



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology

Academic Year 2024– 2025(Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CS3362 & Foundations of Data science

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit 5 / Geographic Data with Basemap
- **Course Outcome:** The student will be able to apply visualization libraries in python to interpret and explore data
- **Topic Learning Outcome:** TLO11.
- **Activity Chosen:** Flipped Classroom
- **Justification:**
 - Geographic data with basemap is an important topic The question was asked in the majority of university questions, and it is one that is important.
 - Students obtain an understanding of leadership and voluntarism through this practice.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - The topic was given to Rasith Meeran of second year IT one week before.
 - The material is also shared to student.
 - On the day of activity ask the students to teach the topic. The topic was delivered around 20 minutes with demonstration given using Google colab.
 - He interacted to the students well, and the students were asked doubt and he clears the doubt.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO12	PSO1	PSO2	PSO3
CO5	3	3	2	2	2	2	3	2

(1 – Low

2 – Moderate

3 – High)



- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO12	PSO1	PSO2	PSO3
	3	3	2	2	2	2	3	2
Justification for correlation	Apply basic Knowledge and fundamentals of data visualization techniques for solving practical problems	Identify the need of data visualization techniques for developing applications	Able to design solutions for applications implementing data visualization techniques to find the relation between data	Able to analyze the relationship between data and provide valid conclusions based on visualization	Ability to recognize the need for visualization techniques in real world applications	Exhibit knowledge in visualization techniques to provide solution for complex computational problems	Exhibit knowledge in visualization techniques in the field of IoT, Cloud Computing and Data Analytics to provide IT solutions.	Exhibit knowledge in visualization techniques in software projects to deliver a quality product



- **Images / Screenshot of the practice:**

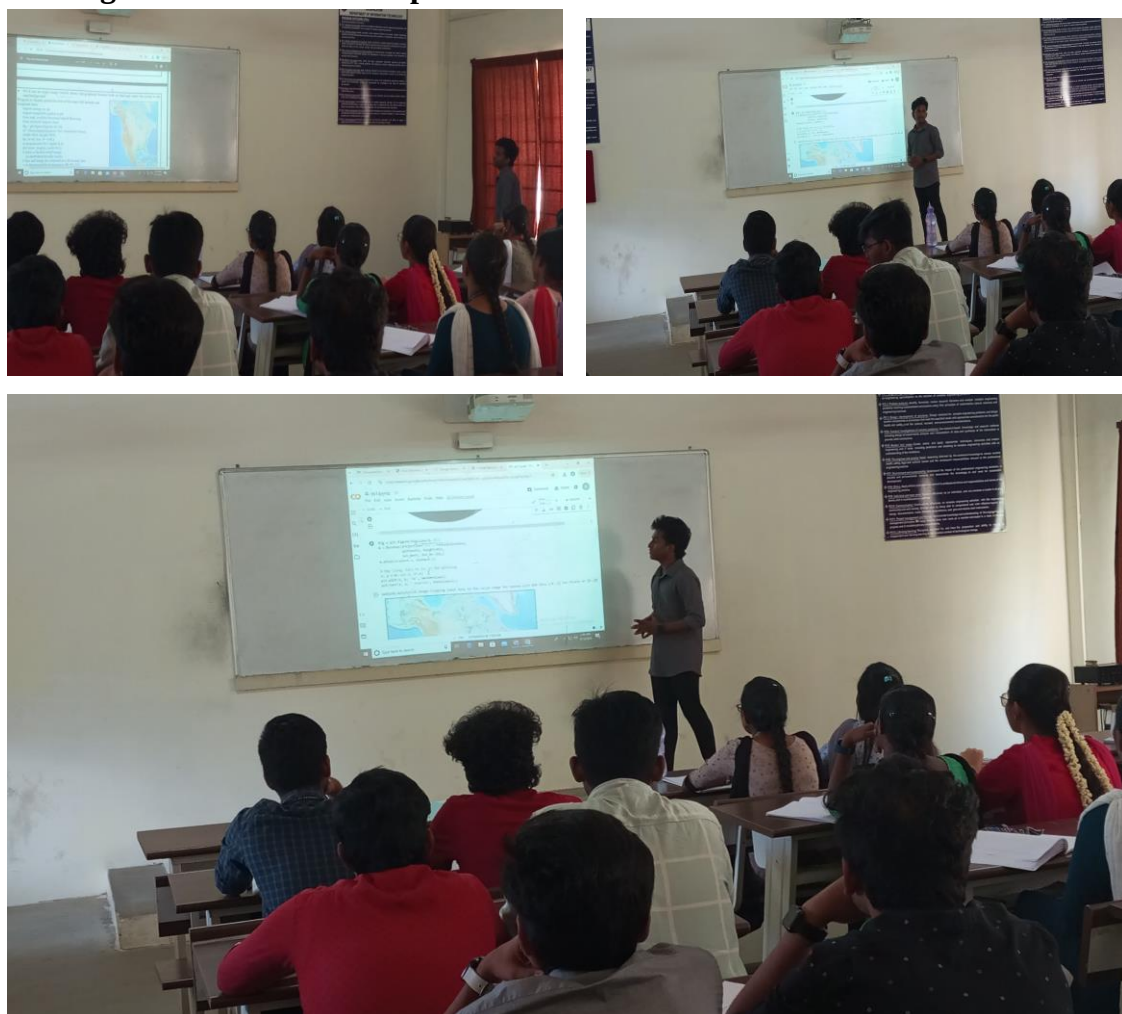


Figure 1: The topic Geographic Data with Basemap by Rasith Meeran

- **Reflective Critique:**

- **Feedback of practice from students and other stakeholders:**

- Everyone in the class paid close attention to the discussion, recognizing the importance of data visualization.
 - Students became familiar with the geographic representation provided by the Basemap package.
 - This activity helped students understand different map projections.

- **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Students learned to identify regions with rivers, coasts, and land by analyzing geographic data.



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- They were able to visualize these regions using various colors and design elements.
- This task helped students recognize the significance of visualization packages.
- **Challenges faced in implementation:**
 - Initially, some students were noisy during the presentation, but after the demonstration using Colab, they became more engaged.
 - The topic was planned for 10 minutes, but it ended up taking 20 minutes.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/>
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024/IP/4_CS3352_VA_LT.pdf

Signature of Faculty Member

HOD